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INTRODUCTION TO INNOVATION

Invention. Creating something new that has never existed before. To be patented an invention needs an **inventive step** (non-obviousness): it must be an adequate distance beyond or above the **state of the art** (what is already known in that field). In plain words: if an expert in the field could easily have thought of it already, it cannot be patented.

Innovation. WIPO: **"Innovation means doing something new that improves a product, process or service."** Kelly and Kranzberg (1978): innovation is the process of **making changes to something established by introducing something new that adds value to customers and contributes to the knowledge store of the organisation. Innovation management** allows the organisation to respond to external or internal opportunities and use its creativity to introduce new ideas, processes or products. In plain words: an invention only becomes an innovation when it is put to use and gives customers value.

Invention vs innovation (main differences). Origin and essence: invention pioneers entirely new concepts; innovation adapts and reimagines existing creations. **Impact and evolution:** invention marks a singular groundbreaking moment; innovation evolves through continuous improvement. **Risk and uncertainty:** invention entails venturing into uncharted territory; innovation involves calculated risk-taking. **Time frame:** invention needs extensive research, development and testing; innovation is a more rapid implementation through refining or adapting existing products. **Parties involved:** invention is led by the research and development department; innovation requires cross-departmental collaboration.

Invention vs innovation, the examples. Microprocessor – invention: Intel's Ted Hoff and Federico Faggin, a small electronic device that is the brain of a computer; innovation: the development and evolution of microprocessors, enabling countless electronic devices, more computing power and the foundation of the digital age. **Zipper** – invention: Gideon Sundback's fastening device; innovation: integrating zippers into clothing, bags and other products, streamlining fastening and enabling new designs. **Xerox** – invention: Xerox PARC (Palo Alto Research Center) invented the computer mouse and the GUI; innovation: Apple and others adopted and developed them, giving widespread GUIs and user-friendly computing. **Telephone** – invention: Alexander Graham Bell, voice communication over long distances; innovation: telephone networks and the global telecommunications industry. **Printing press** – invention: Johannes Gutenberg, mass production of books; innovation: spreading knowledge, democratising information and building publishing industries (education, cultural exchange, spread of ideas). **Steam engine** – invention: key contributions from James Watt; innovation: using it in many applications, advancing manufacturing, transport and the mechanisation of industries. **Internet** – invention: a global network of interconnected computers built by many individuals and organisations; innovation: communication, information sharing, e-commerce, social networking and the digital transformation of society. **Pattern: the invention is the new thing itself; the innovation is what people do with it so it reaches many users.**

Product, service and process innovation. Product = a new or better good (slide example: a shoe). **Service** = a new or better way of serving customers (a food tray served by a waiter). **Process** = a new or better way of making or doing things (a business-process flow; the Ford assembly line below).

Incremental and radical innovation. Radical innovation is about making major changes in something established. Incremental innovations consist of smaller endeavours, making them easier to manage than radical innovation. Graph (Revenue/Efficiency against Time): incremental innovation is a staircase of many small steps up; radical innovation is one big jump up. In plain words: radical = big change, high risk, can create a new market; incremental = small improvements, lower risk, keeps current customers.

Architectural and modular innovation. The slides use a camera taken apart into its parts and a camera built as a modular rig, with a 2 x 2 chart. **Vertical axis:** impact on architectural knowledge (low at the bottom, high at the top). **Horizontal axis:** impact on component knowledge (low at the left, high at the right). **High architectural + high component = Radical. High architectural + low component = Architectural. Low architectural + high component = Modular. Low + low = Incremental.** In plain words: **architectural** = the way the parts are linked together (the system design) changes while the parts stay mostly the same; **modular** = one component is replaced or improved while the overall design stays the same; radical changes both; incremental changes neither much.

Seven sources of innovation (Peter Drucker). 1 The unexpected (unexpected success, unexpected failure, unexpected outside event) · **2 The incongruity** (when the need is incongruent with supply) · **3 Innovation based on process need** (necessity is the mother of innovation) · **4 Changes in industry structure or market structure** that catch everyone unawares · **5 Demographics** (different life-style needs) · **6 Changes in perception, mood and meaning** · **7 New knowledge** (AI, nanotechnology; scientific and non-scientific).

1 The unexpected. Penicillin: the first antibiotic was the accidental by-product of a messy workspace. Alexander Fleming (bacteriologist, London) came back from vacation in 1928 and found mould growing on a Petri dish (unintended contamination) with no bacteria around it; he named it penicillin after the fungus *Penicillium notatum* and published in 1929 but was unsure of a use because it was hard to purify and stabilise. A decade later Oxford chemists turned it into viable medicine: first tested on a patient in 1940, widespread use from 1942, now the most commonly used antibiotic. **Viagra:** developed to treat angina (a heart condition that constricts the vessels supplying the heart); in drug trials it failed to prevent angina but gave another result, more erections in male participants, a disappointment at the time but a gold mine for Pfizer. In plain words: an unexpected result (success, failure or outside event) can open a new product.

2 The incongruity (need incongruent with supply), five types. Between reality and assumptions = a gap between what everyone "knows" or assumes and the actual reality (outdated information, cultural biases, lack of awareness of recent changes). **In a process** = the current way of doing things does not make sense and there seems to be a more effective or efficient way to get the desired result. **Between perception and reality** = people's perception does not match reality (misinformation, misunderstanding, bias), which creates opportunities to correct misconceptions or to create products and services that match people's perceptions. **In industry or market structure** = changes in market share, the competitive landscape or new technologies that disrupt traditional industry structures create discrepancies or imbalances. **Demographic incongruity** = changes in population size, age, ethnicity, gender or other factors create new needs or wants that existing products or services do not meet.

3 Process need. "Necessity is the mother of innovation." Example: the Ford automated assembly car line (slide: workers assembling by hand, the moving assembly line with Model T cars, a modern robotic line). A need inside the process (build cars faster and cheaper) led to the assembly-line innovation.

Sources 4 to 7 explained. 4 Industry or market structure: when the industry or market changes suddenly, new entrants can win (a new platform or technology that disrupts the old players). **5 Demographics:** changes in population (age, size, gender, ethnicity) create new needs, e.g. an ageing population needs health products. **6 Perception, mood and meaning:** when people's attitudes change (e.g. more health awareness) they want different products, e.g. organic food. **7 New knowledge:** new science or technology (AI, nanotechnology) opens new products.

INNOVATION MANAGEMENT TECHNIQUES

What IMTs are. Innovation does not always mean using the very latest cutting-edge technology; it is less a question of technology and more a way of thinking and finding creative solutions within the company. **IMTs (innovation management techniques or tools) are a range of tools, techniques and methodologies that help companies adapt to circumstances and meet market challenges in a systematic way.** They are well developed and standardised, and aim to improve the competitiveness of firms by focusing on knowledge. **The nine IMTs: a Knowledge management, b Intellectual property rights, c Customer relationship management, d Business intelligence, e Brainstorming, f Mind mapping, g Prototyping, h Quality function deployment, i Business simulation.**

a Knowledge management (KM). The process of creating, sharing, using and managing the knowledge and information of an organisation. Perspectives: community of practice, intellectual property. **Dimensions: tacit knowledge and explicit knowledge. Strategies:** knowledge sharing (inter-project, inter- and intra-organisational), storytelling, after-action reviews or reflections, knowledge mapping, CoP (the slide writes it as code of practice), expert systems. **Technologies:** groupware, content and document management systems, e-learning systems. In plain words: **tacit** knowledge is in people's heads and experience and is hard to write down (a chef's feel for taste); **explicit** knowledge is written down and easy to share (manuals, databases, documents). KM tries to turn tacit into explicit so the company does not lose it when people leave.

b Intellectual property rights (IPR) management. IPR is a potential source of market power: managing existing IPRs and developing a culture to create new IPRs is important to an organisation. **Strategic use of IPR: Sell, License, Collaborate, Donate. Patent search or technology search** helps in developing alternatives. In plain words: **sell** = transfer the ownership for money; **license** = let others use it for a fee (royalty) while you keep ownership; **collaborate** = share it with partners to develop something together; **donate** = give it away (to a university or the public) for goodwill or wider adoption.

c Customer relationship management (CRM). Gathering, managing and using customer information is imperative to sustain a business. Customer interviews focusing on customer pain points are the core of product development. Social media and other technologies help businesses connect with their customers and seek their input during the idea-generation stage of the product development life cycle. In plain words: a pain point is a problem the customer faces now; if you find it by asking customers directly, your product solves a real need.

d Business intelligence (BI). BI technologies can handle large amounts of structured and unstructured data to help identify, develop and otherwise create new strategic business opportunities (for your products and for competitor products). **Technologies:** reporting, online analytical processing (OLAP), analytics, data mining, process mining, complex event processing, business performance management, benchmarking, text mining, predictive analytics and prescriptive analytics. In plain words: BI turns a company's raw data into insight for decisions; predictive analytics says what is likely to happen, prescriptive analytics says what to do about it.

e Brainstorming. A group creativity technique to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members. Rules: go for quantity, withhold criticism, welcome wild ideas, combine and improve ideas. Normally followed by **affinity grouping** (collecting similar items together).

f Mind mapping. A diagram for representing tasks, words, concepts or items linked to and arranged around a central concept or subject, using a non-linear graphical layout that lets the user build an intuitive framework around the central concept. It is a visualising technique similar to concept maps and modelling graphs (the slide example is a mind map of Inkscape's features: the central topic in the middle with branches and sub-branches around it).

g Prototyping. A prototype is an early sample, model or release of a product built to test a concept or process. Prototyping provides specifications for a real, working system rather than a theoretical one. **Rapid prototyping** is a group of techniques used to quickly fabricate a scale model of a physical part or assembly using three-dimensional computer-aided design (CAD) data. In plain words: build a cheap early version, let users try it, and fix problems before spending on the final product.

h Quality function deployment (QFD). A method to transform qualitative user demands into quantitative parameters, to deploy the functions forming quality, and to deploy methods for achieving the design quality into subsystems and component parts, and ultimately into specific elements of the manufacturing process. QFD is applied in product design, manufacturing, production, engineering, research and development (R&D), information technology, support, testing, regulatory and other phases, in hardware, software, service and system organisations. In plain words: it converts what customers say they want ("easy to hold") into measurable technical targets (weight in grams); the tool is the House of Quality (Lecture 4).

i Business simulation. Simulation used for business training, education or analysis. Business process simulation is used to assess the dynamic behaviour of processes over time, i.e. how process and resource performance react to changes or fluctuations of certain environment or system parameters; the results give insights on how to improve factors such as process performance, process and product quality, customer satisfaction or resource utilisation. **Marketing simulation games** give participants an interactive way of testing marketing decisions on marketing strategy, product positioning, pricing strategies and consumer behaviour. In plain words: you test decisions on a model first, so mistakes cost nothing.

DYNAMICS OF IDEATION AND CREATIVITY

Idea and ideation. Ideation is the process of generating, developing and communicating new ideas (the formation of ideas and concepts); it can be done by individuals, organisations or crowds. It refers to developing and conveying prescriptive ideas to others, typically in a business setting, and describes the sequence of thoughts from the original concept to the implementation. Ideation can spring from past or present knowledge, external influences, opinions, convictions or principles, and can be expressed in graphical, written or verbal terms. **Creativity** is a phenomenon whereby something somehow new and somehow valuable is formed.

Creativity. The characteristic of a person to generate new ideas, alternatives, solutions and possibilities in a unique and different way; the ability to conceive something unpredictable, original and unique. It must be expressive, exciting and imaginative. It is an activity in which a person has to think beyond imagination to bring something worthwhile, and it is an activity of unveiling something that was previously hidden. In plain words: ideation is the process of producing ideas; creativity is the ability that makes those ideas new and valuable.

Creative process. The evolution of an idea into its final form through a progression of thoughts and actions; it involves critical thinking and problem-solving skills. The stages were first articulated by **Graham Wallas in 1926**. The slide shows **5 stages: 1 Preparation (research, consume, immerse, gather) → 2 Incubation (explore, experiment, synthesise) → 3 Illumination (the "Aha!" moment) → 4 Evaluation (reflect, criticise, assess) → 5 Implementation (work).** Note: Wallas's original model has four stages, Preparation, Incubation, Illumination and **Verification**; verification is the same step as evaluation (testing the idea). In plain words: learn about the problem, step away and let the mind work on it, get the sudden insight, check whether it really works, then build it.

Creative idea generation techniques (10). Role playing: designers act out scenarios. **Active search:** designers hunt for a particular solution. **Attribute list:** take an existing product or system, break it into parts, then recombine these to identify new forms of the product or system. **Brainstorm:** generate a large number of solutions to a problem, focusing on the quantity of ideas. **Critique:** receiving input on current design ideas, collaboratively (a colleague's design critique) or by critiquing your own ideas. **Collaborate:** two or more people working together towards a common goal. **Documenting:** designers write down ideas (physically or electronically). **Expert opinion:** designers elicit opinions from experts to identify potential problems with products or services before more comprehensive evaluations. **Empathy / user research:** the designer observes people in everyday situations to develop empathy for them. **Storyboards:** a way for designers to represent information gained in the research phase of the design process.

How to enhance creativity (8). Providing creativity training (what separates the average person from Edison, Picasso or Shakespeare is not creative capacity but the ability to tap it by encouraging creative impulses and acting on them; training helps everyone tap that capacity). **Providing support** (give employees the tools and resources they need; one of the most valuable is time). **Rewarding creativity** (money, praise, recognition and celebration are powerful incentives). **Modelling creative behaviour** (entrepreneurs who set examples, take chances and challenge the status quo soon find employees doing the same). **Expecting creativity** (give employees permission to be creative). **Expecting and tolerating failure** (creative ideas produce failures as well as successes; managers must remove the fear of failure). **Encouraging curiosity** (constantly asking "what if..." and taking a "maybe we could..." attitude breaks the assumptions that limit creativity). **Viewing problems as challenges** (every problem is an opportunity for innovation; dumping problems on employees' desks to be "fixed" does nothing to develop their creativity). **Slide example, Reddit:** the idea was taken from another social news site, Slashdot, but the upvote/downvote system made it stand out. Creating a whole new product with completely new features may not click with consumers; they relate more when you take features that already exist and deliver more value (like Instagram copying Snapchat's "Stories").

Barriers to creativity (9). Searching for the one "right" answer (schools teach there is one right answer, which blocks brainstorming). **Focusing on "being logical"** (logic is valuable when evaluating and implementing ideas, but too much of it in the early imaginative phase discourages intuition). **Blindly following the rules** (creativity often depends on breaking existing rules to find new ways). **Constantly being practical** (suspending practicality for a while frees the mind to consider solutions that would otherwise never arise). **Becoming overly specialised** (defining a problem as one area of specialty limits seeing how it relates to other issues). **Avoiding ambiguity** (ambiguity pushes us to "think something different" and stretch beyond normal boundaries). **Fearing looking foolish** (new ideas are rarely born in a conforming environment; people conform to avoid looking foolish). **Fearing mistakes and failure** (failure is not an end but a pit stop on the way to success). **Believing "I'm not creative"** (a person who believes this behaves that way and makes it true; everyone has the potential, but must tap into it first).

Opportunity. A favourable set of circumstances that creates a need for a new product, service or process. From an economic perspective, entrepreneurial opportunities generally refer to situations that hold the potential for new economic value. Opportunities exist in reality; the role of the entrepreneur is to identify them and the ones that hold the potential to generate economic value.

The process: turning an idea into an opportunity. 1 Strategic fit (you need something the market wants, or the market needs to want what you have) · **2 Business plan** (needed to convert an idea into an opportunity) · **3 Team** (needed to deliver on an opportunity) · **4 Leadership** (guide your team effectively) · **5 Resources** (needed to turn the idea into an opportunity). Successful entrepreneurs are good at figuring out which ideas can be turned into opportunities, and then they turn them into one.

Inbound ideation (internal innovation ideation). Ideation done inside the company, a conventional approach used for decades in the 20th century. It can be a brainstorming or rapid-ideation session with 10 or 20 people in a limited time frame, or web-based ideation tools that bring in thousands of participants from around the world to share ideas in real time. It works best by leveraging the collective brainpower of the entire company, where every employee raises their potential and feels a sense of purpose; companies offer incentives for the best ideas or reward top contributors. **Key elements to make it effective: (a) proper framing of the challenge statement, (b) high levels of engagement and contribution, (c) making rewards and recognition part of the process, (d) publishing the results of ideas accepted and implemented, (e) an effective tool kit to analyse and interpret.**

Outbound ideation. The marketplace is the final judge of business performance, so the environment outside the organisation plays a vital role in ideation. Methods: voice of customer, open innovation systems and other miscellaneous methods.

Voice of the customer. The dynamics of the market give cues for correct innovation; a good innovation design should come from the customer's needs and an understanding of what drives customer satisfaction. **Ethnographic method:** create the product or service from a deep understanding of the customer's **unarticulated needs** (Dorothy Leonard-Barton) by observing the customer over a longer period, even camping with them; it gives the greatest insight into customer behaviour so products are modified. Observation techniques: **discrete observation** (customers are observed without their knowledge or permission; useful for consumer products and services used in public, e.g. a car maker), **contextual interviews** (cover substantial background information that observation alone does not give; asking and probing uncovers customer requirements, perception and product usage), **empathy building** (merging yourself with the customer, being the customer, to get an overall perspective). **Customer visit teams:** cross-functional teams from various departments visit customers and use semi-structured interviews to gather information and data. **Customer focus groups:** used for need assessment, need gaps, problem identification, issues to address and suggestions for improving the product; the focus is specific and defined, so it gives a need-based perspective. **Lead users: groups of customers or users that face more challenging requirements than most of the current marketplace; their needs anticipate the general need of the market by months or years.** They are intelligent, imaginative customers with a futuristic perspective; companies identify them and work closely with them, selecting them from different market segments including current users and analogous users from other sectors. **Screening the existing users:** identify customers who are more demanding than others, the extreme users, who are extreme in how they use and treat the product, challenge it, and are experts about it. **Customer or user design:** users are invited into a simulated IT modelling set-up to design, modify and innovate the product (GE gives customers web-based tools to design better plastic products). **Customer brainstorming:** brainstorming with selected customers on product innovation at customer events in B2B and B2C businesses. **Customer advisory panel:** panels or boards of experts and specialists who interact with customers on product development.

Open innovation ideation. Methods where companies explore ideation outside their organisation: 1 External idea contest (events where outsiders are invited to submit ideas) · **2 Partnering with vendors** (collaborating with vendors in forward and backward integration for idea generation) · **3 Scanning small businesses and start-ups** (they are rooted at the grass roots of the market and are great reservoirs of ideas) · **4 Accessing external know-how** (the external scientific and technical community; online tools such as NineSigma and Yet2.com help) · **5 External product design** (the company seeks not only ideas but complete product designs from all business stakeholders).

Other miscellaneous ideation methods (3). Peripheral vision: rigorously studying the environment formally and informally, identifying gaps and links, and then developing a vision. **Disruptive technology:** disruption of technologies opens new vistas for innovation; it needs foresight to tap into the new technologies and fill the gap opened by the disruption of the existing technology. **Patent mapping:** mapping and mining existing patents shows where competitive activity is located, which indicates potential areas for new product development.

NEW PRODUCT DEVELOPMENT PROCESS

New product development (NPD). The complete process of bringing a new product to market, renewing an existing product or introducing a product in a new market; the transformation of a market opportunity into a product available for sale. The product can be tangible or intangible (a service, experience or belief). NPD requires an understanding of customer needs and wants, the competitive environment and the nature of the market. **Cost, time and quality** are the main variables that drive customer needs. Innovative companies develop continuous practices and strategies to better satisfy customer requirements and increase their market share by regularly developing new products. **Product development** (also called new product management) is a series of steps that includes the conceptualisation, design, development and marketing of newly created or rebranded goods and services; it covers a product's entire journey from the initial idea to its market release and later.

How to create a product development plan (5 steps). 1 Identify a product need and business case (test marketing and surveys gauge interest, so the product has a strong reason to exist given its business model). **2 Create a product vision** (the purpose of the product, what it does, who it is for and what the design is). **3 Draft a roadmap** (a detailed plan of action showing what goals should be met and when). **4 Implement the roadmap** (product iterations are made, reviewed and improved). **5 Continue with development and assessments** (teams work on enhancements and changes and gather customer feedback to improve the product).

NPD process: 8 stages. 1 Idea generation, 2 Idea screening, 3 Concept development and testing, 4 Market strategy and business analysis, 5 Feasibility study, 6 Technical design and roadmap, 7 Test marketing, 8 Market entry and commercialisation. 1 Idea generation: the continuous and systematic quest for new opportunities, including updating or changing an existing product; the goal is ideas for new products or services, or improvements, that address a gap in the market. **2 Idea screening:** weeds out flawed or less desirable ideas through objective consideration, early testing and feedback from consumers and other stakeholders. **3 Concept development and testing:** the product concept and functionality are tested on a customer base and developed further from the feedback (in car manufacturing, developers make concept vehicles out of clay and show them at auto shows to get consumer feedback). **4 Market strategy and business analysis:** the optimal way to market and sell the product is determined (it can differ for a new market or an existing one); the market strategy is made of the **four Ps: product** (a service or good designed to satisfy the demand of a target audience), **price** (pricing affects profit margins, supply and demand and market strategy), **promotion** (presenting the product to the target audience to increase demand and show its value proposition: advertisements, public relations, marketing campaigns) and **place** (brick-and-mortar, online-only or omnichannel; in the digital economy customers are often engaged and converted online). **5 Feasibility study** asks: do you have the labour and materials required? what is the cost of production, delivery and promotion? do you have access to the right distribution channels? **6 Technical design and roadmap:** integrates the feasibility results and feedback into the product, turns prototypes and concepts into workable market offerings, irons out technical difficulties and prepares business groups (R&D, finance, marketing, production, operations) for the launch; it also creates the product roadmap the teams will follow. **7 Test marketing:** validates the entire concept, from marketing angles to packaging, advertising and distribution, usually by offering the product to a random sample of the target market, so the reception can be reviewed before a full go-to-market strategy. **8 Market entry and commercialisation:** the product is introduced to the target market and made available to everyone, and the product life cycle begins.

Real-world example, Uber. Idea generation: in 2008 Travis Kalanick and Garrett Camp discussed using a smartphone app to solve the difficulty of getting a taxi on a snowy evening in Paris (smartphones were then in their infancy). **Product technical design:** Uber's engineers used well-known languages such as Python, Node.js and Java. **Concept testing:** in 2010 the first Uber ride was requested and taken in San Francisco. **Market entry:** in December 2011 the app was rolled out internationally, beginning in Paris.

Design thinking (a product development framework), 5 steps. 1 Empathise (learn more about a problem from multiple perspectives; learn about the audience you are designing for by observation and interview: who is my user, what matters to this person?) → **2 Define** (identify the scope and true nature of the problem; create a point of view based on user needs and insights: what are their needs?) → **3 Ideate** (brainstorm as many creative solutions as possible; wild ideas are encouraged) → **4 Prototype** (build a representation of one or more ideas to show others and weed out unworkable or impractical solutions; a prototype is just a rough draft) → **5 Test** (solicit feedback by sharing the prototype with the original user: what worked, what did not?).

Stage-gate process. A typical stage-gate process runs Discover → Scope → Design → Develop → Scale up → Launch, with a gate (decision point) between stages, internal ideation programmes and external idea networks feeding in, and continuous customer or user interface throughout. **Each stage is designed to collect specific information to move the project to the next stage or decision point; information gathered in each stage is used to reduce levels of technical uncertainty and economic risk; each gate has a unique set of quantitative and/or qualitative criteria for deciding whether to fund or approve the project into the next stage. Slide's 5-stage version (stage / gate):** 1 Preliminary investigation and analysis (scoping studies, technical and market assessments, idea generation) / Gate 1 Research project selection; 2 Concept definition (early research to explore and define the technical concept) / Gate 2 Research approval; 3 Concept development (develop and test a prototype, simulation, scalability and end-user acceptability) / Gate 3 Proof of technical feasibility; 4 Technology development and verification (pilot-scale development, verification of technical performance, validation of economic potential in field tests) / Gate 4 Proof of commercial feasibility; 5 Information dissemination and commercialisation (commercial launch, production-scale manufacture, developing market infrastructure). **Gate review decisions: Go forward** (goals of the previous stage were met and funding is approved), **Stop** (the project is not progressing, the market shifted permanently, the technology is obsolete or objectives cannot be met), **Hold** (suspended for a period; criteria re-evaluated later), **Return** (goals not met but the project is still a high priority, so the team returns to the previous stage to complete the work).

House of quality (HoQ). The basic design tool of quality function deployment (QFD); it originated in 1972 at Mitsubishi's Kobe shipyard. It has been used by Japanese manufacturers of consumer electronics, home appliances, clothing, integrated circuits, synthetic rubber, construction equipment and agricultural engines, and for services such as swimming schools, retail outlets and even apartment layouts. It is a kind of **conceptual map** that provides the means for inter-functional planning and communication. **Steps:** (1) it begins with the customer, whose requirements are called **customer attributes (CA)**, usually written in the customers' own words; (2) it measures the **relative importance** to the customer of all CAs (weightings based on team members' direct experience with customers or on surveys); (3) check whether delivering the perceived needs gives a competitive edge: comparison with competitors identifies opportunities for improvement, and perceptual maps based on bundles of CAs identify strategic positioning; (4) list the **engineering characteristics (EC)** that describe the product in measurable terms and directly affect customer perceptions (one EC may affect more than one CA; this is a time for systematic, patient analysis and brainstorming); (5) the **relationship matrix** shows how much each EC affects each CA, using expert engineering experience, customer responses and tabulated data from statistical studies or controlled experiments, with numbers or symbols for the strength of the relationships; (6) when objective measures are known, the team sets **target values**, the ideal new measures for each EC in the redesigned product; (7) the distinctive **roof matrix** helps engineers specify the engineering features that must be improved together and facilitates engineering trade-offs. The HoQ is custom-made to the requirements; design teams may add columns for factors relevant to new developments. Its principles apply to any effort to establish clear relations between manufacturing functions and customer satisfaction. **Four linked houses carry the voice of the customer through to manufacturing:** I House of quality (customer attributes → engineering characteristics), II Parts deployment (engineering characteristics → parts characteristics), III Process planning (parts characteristics → key process operations), IV Production planning (key process operations → production requirements).

Innovation sweet spot (IDEO). The ideal innovation process is the trifecta of desirability, feasibility and viability, an idea that originated from IDEO in the early 2000s; an idea that meets all three has a desirable solution (one your customer needs), a feasible solution (builds on the strengths of your current operational capabilities) and a viable solution (a sustainable business model; does it give you profit?). The overlap of the three circles is the **innovation sweet spot**. **Desirability: are we solving the right pain point?** It tests whether the solution is a nice-to-have or a must-have for the customer. Ask: what task am I helping my customer complete, and what does successful completion look like for them? If you solve the key pain points they meet doing that task you pass; if other pain points remain, pivoting may put you on a better path. *Desirability tests whether your innovation is solving the right customer problem.* **Feasibility: are we building on our core operational strengths?** It measures the operational capabilities used in a new solution, asking the organisation to assess its strengths objectively (technology, financial, branding, customer service, partnerships and so on): how can we build this solution to make our business healthier and stronger? If a solution needs completely new capabilities, the investment is riskier because of the resources required and because it may change the market's perception of the company, not always for the better. *Feasibility tests whether your company has enough technical requirements and resources to produce the product.* **Viability: does our solution contribute to long-term growth?** It focuses on the value chain to make sure the solution is viable now and in the future: does our business model fit the way customers want to use and pay for the solution, and is the way we build it and buy from suppliers profitable? Viability looks at profit and also at sustainability, so the business contributes to community and society. *Viability tests whether your company can make a profit out of this product, process or service.* **Sustainable innovation** (the slide adds sustainability as a fourth circle) = developing and implementing new products, processes or services that contribute positively **economically, ecologically and socially**: ecologically it should help reduce resource consumption; socially it should fulfil social needs, such as improving quality of life for all members of society or promoting equal opportunities.

ENTREPRENEUR AND ENTREPRENEURSHIP

Who is an entrepreneur? (Stevenson). Entrepreneurship is the pursuit of opportunity beyond resources controlled – Professor Howard Stevenson. "Pursuit" implies a singular, relentless focus; entrepreneurs often perceive a short window of opportunity, and they need to show tangible progress to attract resources because the mere passage of time consumes limited cash balances. **"Opportunity"** implies an offering that is novel in one or more of four ways: **1 pioneering a truly innovative product; 2 devising a new business model; 3 creating a better or cheaper version of an existing product; 4 targeting an existing product to new sets of customers** (these types are not mutually exclusive). **"Beyond resources controlled"** implies resource constraints: at the outset the founders control only their own human, social and financial capital. Many entrepreneurs **bootstrap**: they keep expenditure to a bare minimum, investing only their own time and, as necessary, personal funds; sometimes this is enough to make the venture self-sustaining from internally generated cash flow. With most high-potential ventures, founders must mobilise more resources than they control personally (production facilities, distribution channels, working capital and so forth).

Four types of risk an entrepreneur faces. Demand risk (prospective customers' willingness to adopt the envisioned solution) · **Technology risk** (high when engineering or scientific breakthroughs are needed to bring the solution to fruition) · **Execution risk** (the ability to attract employees and partners who can implement the plans) · **Financing risk** (whether external capital will be available on reasonable terms). The entrepreneur's task is to manage this uncertainty, recognising that some risks cannot be influenced by their actions.

Entrepreneur and entrepreneurship. An entrepreneur is someone who perceives an opportunity and creates an organisation to pursue it. An entrepreneur is one who creates his own business, i.e. a person who organises, operates and assumes the risk of a business venture. **Entrepreneurship** is the process of action an entrepreneur undertakes to establish the enterprise; the entrepreneur is essentially a business leader and the functions he performs are entrepreneurship.

Characteristics of an entrepreneur. A highly achievement-oriented, enthusiastic and energetic individual who: 1 is action-oriented and highly motivated and takes risks to achieve goals; 2 has unwavering determination and commitment, is creative and result-oriented, and works hard for personal and financial rewards; 3 accepts responsibility with enthusiasm and endurance; 4 has self-confidence, is dedicated, sets self-determined goals and markets ideas in response to the existing market; 5 is a thinker and doer, planner and worker; 6 can foresee the future, with a salesman's persuasiveness, a financial talent for manipulating funds and an auditor's precision; 7 depends on the intelligence, imagination and strength of purpose of the individual. **Qualities table (slide): essential** (foresight, imagination, communication, self-knowledge, autonomy, search skills, analytical ability, networking skills, manage risk, industry knowledge); **rare** (foresight, imagination, autonomy, analytical ability, delegation, organisational skills); **difficult to screen for** (foresight, imagination, self-knowledge, autonomy, search skills, delegation, organisational skills); **observable over time** (all except imagination); **capable of development** (all except imagination).

Why be an entrepreneur: wage employment vs entrepreneurial employment. Nature: self-satiating (wage) vs self-generating (entrepreneurial). **Scope:** limited vs unlimited. **Orientation:** routine, status quo, problem avoiding, dependent vs creative, innovative, problem solving, independent decisions. **Contribution:** consumes national wealth vs generates national wealth. **Earning:** fixed (subsistence) vs growing (generating surplus). **Advantages of being an entrepreneur (word cloud):** independence, achievement, satisfaction, recognition, confidence, learning, personal growth, financial independence, freedom, freedom to innovate, creativity, knowledge, flexibility and work-life balance, happiness, network, customer satisfaction, being your own boss, providing employment, respect.

Skills of an innovative entrepreneur (7). Basic financial skills (budgeting and financial statement analysis; a reasonable budget you stick to can be the difference between success and failure; avoids overspending and allocates resources properly). **Networking** (your network is one of your greatest assets: meet like-minded professionals, build your future team, keep a finger on the pulse of your industry; co-workers, clients, business professionals, industry leaders and speakers, friends and family). **Ability to accept and act on feedback** (stay humble; your idea of the perfect product may not resonate with customers). **Pattern recognition** (in data, market trends and user behaviour; e.g. patterns in cash-flow statements, seasonality in sales data). **Strategic thinking** (critical thinking to solve complex problems and plan for the future; develop analytical, communication, problem-solving, planning and management skills). **Negotiation** (with investors, early employees and partners). **Growth mindset** (intelligence, abilities and talents are learnable and improvable, as opposed to a fixed mindset that sees them as stable and unchangeable).

Classification by ownership (4). Founders or "pure entrepreneurs" conceptualise the business plan and make it a success (Dhirubhai Ambani, Reliance). **Second-generation operators of family-owned business** inherit it (Mukesh and Anil Ambani). **Franchisees:** the parent owner (franchiser) licenses trademarks and tried and proven methods to the franchisee in exchange for a recurring payment; the franchisee did not conceptualise the business but invests money and time. **Owner-managers** buy a business from the founder and then put their time and resources into it (Sabeer Bhatia founded Hotmail; Bill Gates became its owner-manager by buying it for $400 million).

By personality traits and style of running the business. The achiever: a personal desire to excel and make a mark; self-driven, determined whatever the hurdles, a go-getter who often builds the business around a personal brand (Narayana Murthy quit a settled job to start Infosys). **The induced entrepreneur:** induced by external factors such as supportive government policy, unemployment, family support or institutional support; more realistic in approach (entrepreneurs who started small-scale industries when the government announced subsidies and tax rebates). **The idea generator:** highly creative, always searching for innovative ideas, senses demand ahead of others, enjoys first-mover advantage and skims higher profits, the “essence of entrepreneurship” (Bill Gates began making microcomputer software). **The real manager:** runs the business systematically, analyses situations, assesses future opportunities and threats and acts on them; believes in incremental rather than radical change. **The real achievers:** full of life, seeking to achieve not only their own goals but also those of employees, suppliers and distributors.

By type of business (5). **Business entrepreneurs** conceive an idea for a new product or service and combine resources to set up a big or small unit (readymade garments, confectionery; common in trading and manufacturing, and flourish when the business is small). **Industrial entrepreneur:** a manufacturer who identifies customers' potential needs and tailors the product to the market (electronics, textile units, machine tools). **Trading entrepreneur:** undertakes trading and is not concerned with manufacturing; identifies markets, stimulates demand and creates desire among buyers. **Corporate entrepreneur:** shows innovative skill in organising and managing a corporate undertaking registered under a statute (a trust under the trust act, a company under the company act). **Agricultural entrepreneur:** produces and markets crops, fertilisers and other agricultural inputs. **By use of technology: technical entrepreneur** is a “craftsman” type who develops new and improved goods through craftsmanship, concentrates on production more than marketing and is strongest in production techniques; **non-technical entrepreneur** is not concerned with technical aspects and only develops alternative marketing and distribution strategies.

By motivation (4). **Pure entrepreneur:** motivated by psychological and economic rewards, works for personal satisfaction, ego or status. **Induced entrepreneur:** induced by government policy measures giving assistance, incentives and concessions. **Motivated entrepreneur:** motivated by the desire for self-fulfilment. **Spontaneous entrepreneur:** has initiative, boldness and confidence in their ability and starts a business from natural talents. **By profession (3).** **Professional entrepreneurs** are keen on establishing enterprises but sell them once established and move on to start new ones, never clinging to them. **Forced entrepreneurs** are compelled by socio-economic background (jobless or low education). **Individual entrepreneurs** take up activities at individual level, large or small scale, usually in the small-scale sector.

By technical experience (Jones-Evans, 1995). Based on the previous occupational background of entrepreneurs in the technology sector; a technical entrepreneur is the founder and current owner-manager of a technology-based business, primarily responsible for its planning and establishment and with some management control. **Research technical entrepreneur:** did technological research at an academic institution or laboratory before creating a venture (Prof. Amar Bose of MIT, acoustic research, then Bose Corporation). **Producer technical entrepreneur:** a history of direct commercial production or development of a product or process, usually in a large organisation (Subroto Bagchi, Wipro's global R&D chief, co-founded MindTree in 1999). **User technical entrepreneur:** was an end-user of a product or technology (perhaps in support services) without developing it (Tulsi Tanti installed two wind turbines because of power shortages in his textile business and then founded Suzlon). **Opportunist technical entrepreneur:** identified a technology-based opportunity while having little or no technical experience or previous non-technical experience (Azim Premji inherited Wipro from his oil-business father and moved into IT).

By entrepreneurial experience. **Novice entrepreneur:** no prior business ownership experience but now owns an equity stake in a new, purchased or inherited independent business (Narayana Murthy when he founded Infosys on 2 July 1981 with six colleagues). **Habitual entrepreneurs:** experience owning at least two different firms, either one after another or at the same time. **Serial entrepreneur:** has sold or closed a business and now has a stake in a single independent business; thrives on the psychological reward of making an impact (Ryan, 2000), is a risk taker with enough wealth, invests in new ventures that are often vague visions of an unsolved problem, sees failure as experience, and may hand a business to others to manage and return to the start-up stage (Sunil Bharti Mittal, Bharti Airtel). **Portfolio entrepreneur:** currently has stakes in two or more independent businesses at the same time (Kishore Biyani owned Big Bazaar, Pantaloons and Central at once).

Clarence Danhof's classification (4). **1 Aggressive or innovative entrepreneur:** assembles and synthesises information and introduces new combinations of factors of production; senses opportunities for new ideas, technologies, markets and organisations; brings a transformation in life style, very helpful to the country. **2 Imitative or adoptive entrepreneur:** simply adopts innovations introduced by others and sets up the enterprise the same way; very helpful in less developed countries, contributes to growth of enterprise and entrepreneurial culture and, by adopting tested technology, creates employment and acts as an agent of economic development. **3 Fabian entrepreneur:** timid and cautious, imitates other innovations only if certain that failing to do so may damage the business; sceptical, not adaptable to a changing environment, loves old techniques and adopts new technology only when failing to do so would mean loss or collapse, deciding at the last moment. **4 Drone entrepreneur:** conservative or orthodox in outlook; will not give up traditional business, machinery or systems and is comfortable with old-fashioned technology even though the environment and society have changed; a laggard who resists change and refuses to adopt it even at the risk of reduced returns.

Arthur H. Cole classification. Cole divided the history of American entrepreneurship into three stages of entrepreneurial thought: **Empirical** (from scratch to 1860; rarely introduces anything radical or innovative; guided by the rule of thumb), **Rational** (1860 to 1890; has information and is aware of the business environment, understands prevailing economic conditions and introduces modifications when necessary) and **Cognitive** (from 1890 onwards; sophisticated, has experts to advise and acts on the advice, introduces changes whenever necessary, which may completely break from the existing scheme).

Other types. **Spiritual entrepreneur:** e.g. Sri Sri Ravi Shankar, whose Art of Living foundation is present in over 140 countries with over 20 million people having taken its programmes; success lies in the efficiency (usefulness) of the product and in having a philosophy that you “walk” as well as “talk”. **Social entrepreneur:** innovative, highly motivated, creative thinkers combined with the drive to solve social problems; individuals with innovative solutions to society's most pressing problems who are ambitious, persistent, tackle major social issues and offer new ideas for wide-scale change (Florence Nightingale, Maria Montessori, Muhammad Yunus, 2006 Nobel Peace Prize, Vinoba Bhave, Satyam Mishra); they are not content just to give a fish or teach how to fish, but will not rest until they have revolutionised the fishing industry.

Intrapreneurship. **The act of behaving like an entrepreneur while working within a large organisation;** a corporate management style that integrates risk-taking and innovation approaches, and the reward and motivation techniques traditionally associated with entrepreneurship. It refers to employee initiatives in organisations to undertake something new without being asked to do so. The intrapreneur focuses on innovation and creativity and transforms an idea into a profitable venture while operating within the organisational environment.

VALUE PROPOSITION CANVAS AND BUSINESS MODEL CANVAS

Value proposition canvas (VPC): what it is for. **Understanding the customer, with their needs and expectations; developing a product in accordance with what your customer needs and wants; comparing a product you already have with the user's need; finding your product-market fit; avoiding producing something nobody wants.** In plain words (the slides give only these purposes): the VPC has two sides. The **customer profile** lists the customer's **jobs** (what they are trying to get done), **pains** (what annoys, costs or blocks them) and **gains** (what they want or would be delighted by). The **value map** lists your **products and services**, the **pain relievers** (how they remove those pains) and the **gain creators** (how they create those gains). When the value map matches the customer profile you have **fit** (product-market fit).

What is a business model? **A business model describes the rationale of how an organisation creates, delivers and captures value. A business model includes nine basic building blocks** (the business model canvas, from businessmodelgeneration.com). **Layout:** in the middle is the value proposition; on its right customer relationships and channels lead to customer segments; on its left key activities and key resources and then key partners; along the bottom are cost structure (left) and revenue streams (right). In plain words: the right side is the customer side (who and how you reach them), the left side is the infrastructure (what you need to deliver), and the bottom shows the money: what it costs and what it earns.

The nine building blocks and the questions each one asks. **1 Customer segments:** for whom are we creating value? who are our most important customers? **2 Value propositions:** what value do we deliver to the customer? which one of our customer's problems are we helping to solve? which customer needs are we satisfying? what bundles of products and services are we offering to each customer segment? **3 Channels:** through which channels do our customer segments want to be reached? how are we reaching them now? how are our channels integrated? which ones work best? which are most cost-efficient? how are we integrating them with customer routines? **4 Customer relationships:** what type of relationship does each of our customer segments expect us to establish and maintain? which ones have we established? how costly are they? how are they integrated with the rest of our business model? **5 Revenue streams:** for what value are our customers really willing to pay? for what do they currently pay? how are they currently paying? how would they prefer to pay? how much does each revenue stream contribute to overall revenues? **6 Key resources:** what key resources do our value propositions require? our distribution channels? customer relationships? revenue streams? **7 Key activities:** what key activities do our value propositions require? our distribution channels? customer relationships? revenue streams? **8 Key partners:** who are our key partners? who are our key suppliers? which key resources are we acquiring from partners? which key activities do partners perform? **9 Cost structure:** what are the most important costs inherent in our business model? which key resources are most expensive? which key activities are most expensive?

The building blocks in plain words (with examples). **Customer segments:** the groups of people or organisations you serve (students, busy parents, small shops). **Value proposition:** the problem you solve or need you satisfy and the products and services that do it (cheaper rides, fresh organic vegetables delivered). **Channels:** how you communicate with and deliver to customers (website, mobile app, shop, delivery rider, social media). **Customer relationships:** the kind of relationship you keep with each segment (self-service, personal assistance, a community, automated service). **Revenue streams:** how the money comes in (product sales, subscriptions, commission, fees, advertising). **Key resources:** the most important assets you need (people, technology, equipment, brand, money). **Key activities:** the most important things you must do (production, software development, marketing, delivery). **Key partners:** suppliers and partners who help you and whom you rely on. **Cost structure:** the main costs of running the model (salaries, rent, technology, marketing, supplies). Tip: every block must connect to the value proposition, and the revenue should be larger than the costs.

Love your problem. The slides ask “Do you go back?” (comparing an old wristwatch with a smartphone) and give the message **“Love your problem”** with the Google Glass project (a wearable augmented-reality computer with a head-mounted display, camera, bone-conduction audio and voice commands, sold to early adopters at about $1,500). In plain words: fall in love with the customer's problem, not with your solution or technology; a clever product that does not solve a real problem people care about fails (Google Glass).

Desirability, feasibility and viability. The innovation process diagram shows three overlapping areas: **Human values (usability, desirability)**, **Technology (feasibility)** and **Business (viability)**; the centre where all three meet is the innovation or idea sweet spot (the same idea as in Lecture 4, from IDEO). **Customer factory model.** **Every business has a common universal goal: making happy customers.** The customer factory has five stages: **1 Acquisition, 2 Activation, 3 Retention, 4 Revenue, 5 Referral** (customers come in, have a first good experience, come back, pay, and tell others). **Traction modelling.** A traction model sets milestones for growth over about 36 months: problem/solution fit, year 1, product/market fit and then scale, with a minimum success criterion (the slide example is $10 million revenue in 3 years) and a 10X growth model. In plain words: traction is evidence that customers really want the product (users, sales, growth), and the model plans how fast you must grow to be a worthwhile business.

BUSINESS PLAN AND INVESTORS

Why a business plan is important. **Use it as a road map for your business; to show your bank manager; to show your investors and employees** (we need to be on the correct path to reach the destination we want). **First:** you should state the purpose of your business clearly: what is the question to which my business is the answer? **Second:** the plan must be clear for investors to invest in your business; this helps to collect **soft money, bank lending and investor funding.**

What is my business? (business idea and justification). Be clear: customers must know why they should give you money. Write a short but complete description of the business and where you want to take it; this gives focus and clarity for managing it. Example: “helping people with computers” is not wrong but is too vague to manage or to attract customers; “we will provide computer support to small businesses (up to ten staff) in my city” is much clearer. The short statement (about 30 words) is also what you say when someone asks “so, what do you do?”: what you do and why they should use you.

Mission: our reason for being. The reason your business exists, summarised in the **mission statement**. It must include **the purpose of the business; why it needs to exist (the question to which the business is the answer); what it does (sells food, offers advice, supports disadvantaged people); the values you believe in and will run the business by.** It must appeal to all stakeholders: individuals and groups with an interest in the business and how it is managed (shareholders, funders, customers, suppliers, trade unions, pressure groups, the media). It should be short, one to three sentences (about 30 words). Examples: **Toyota** – “to create a more prosperous society through automobile manufacturing”; **Google** – “to make the world's information universally accessible and useful”; **Starbucks** – “to inspire and nurture the human spirit, one person, one cup and one neighbourhood at a time”. Small building business: “our job is to help local people make their home a nicer place to live; we will do this with honesty and never leave you to work on another job.”

Vision: where will you take your business? **Mission is based on the now (what we do and how we do it); vision is focused on the future (where we want to take the business).** A vision statement matters because it shows the scale of our ambition; funders want to know when they will get their money back, so the future of the business is important; it is a future goal that we and our staff can work towards. It must paint a picture of the business and the future world. Examples: **Toyota** – to be the most respected and admired company; **Google** – to provide access to the world's information in one click; **Starbucks** – to establish Starbucks as the premier purveyor of the finest coffee in the world while maintaining uncompromising principles as we grow; a takeaway business – “to be the most recognised national chain of Chinese takeaway restaurants in Sri Lanka by 2030”; a law firm – to be the foremost authority on international company law and international business.

Executive summary. A summary of the key points of the plan (what the business is, how it will make money, strength of its competitive advantage) so readers can decide whether to read on. **It is written last but presented first,** because it needs the information from the main sections. **Its only goal is to get the reader interested in our business without reading the full plan.** **It must answer five crucial questions:** **1 Is there an attractive market? 2 Does the business have the products or services to win? 3 Does the business have the people to make it happen? 4 How much money will the business make? 5 Where will the business go?** **It should be:** in positive language; start with a bold statement and end with a bold statement; focused on the five questions; a synopsis of the whole plan; use numbers to substantiate points; drawn only from the main part of the plan (no new or different information, which would confuse the reader).

What each of the five questions covers. **1 Attractive market:** start with a bold statement (e.g. government incentives are creating a market growing at 30% a year with a shortage of firms, so turnover of £1m in two years is possible; a major retailer confirms an unfulfilled need and would order 1,000 units); then move from general to specific (e.g. for care of older people: population growth figures, government policy shifting from hospital to home care, published data that older people prefer home care, figures for the financial size of the market). **2 Products or services to win:** what is our product or service for this market? show with evidence why it works; present market-research data on the positive response of target segments; who are the competitors? and critically why will ours be better? **3 People:** who will run the business? name the key individuals; their relevant experience; why they are good for the business; how committed they are to staying. **4 Money:** a financial summary with expected income in the first year (market size and likely share), turnover for years 1, 2 and 3, gross and net profit expectations for each year, the repayment schedule for debt, results of sensitivity analysis and return on investment for investors. **5 Where will the business go:** how will you keep growing (new products for this customer group, existing products for new customers; organically or by merger or acquisition), where will the business be in five years, and your personal plans (let it grow forever, or grow it to a certain point and sell it).

Objectives. **Written statements that define the results you want your company to achieve and detail how and when it will achieve them.** They typically focus on key areas such as growth, revenue, productivity and operational efficiency that bring you closer to your long-term vision, and keep you and your employees motivated and focused. **The most effective objectives are SMART: specific, measurable, achievable, relevant and time-bound.** Example 1 (revenue): increase overall revenue by 20% this year by introducing a customer referral programme and sales bonuses in Q1, surveying how targeted customers discovered the brand, and raising marketing spend in those areas by 25%. Example 2 (brand awareness): double Instagram and/or TikTok followers by the end of the year by hiring a full-time social media manager and dedicating an extra 10% of marketing money to social media campaigns and advertising.

Values and the triple bottom line. Values the business must consider include issues such as the use of child labour in manufacturing, deforestation, inequality of wealth and the power of very large corporations. The **triple bottom line** says judging a business by financial performance only is no longer adequate: add a **social** bottom line (impact on people and society) and an **environmental** bottom line (impact on the environment) to the financial one (profit, people, planet). It does not mean abandoning profit, because without profit there is no sustainability, but taking a more balanced approach to company performance.

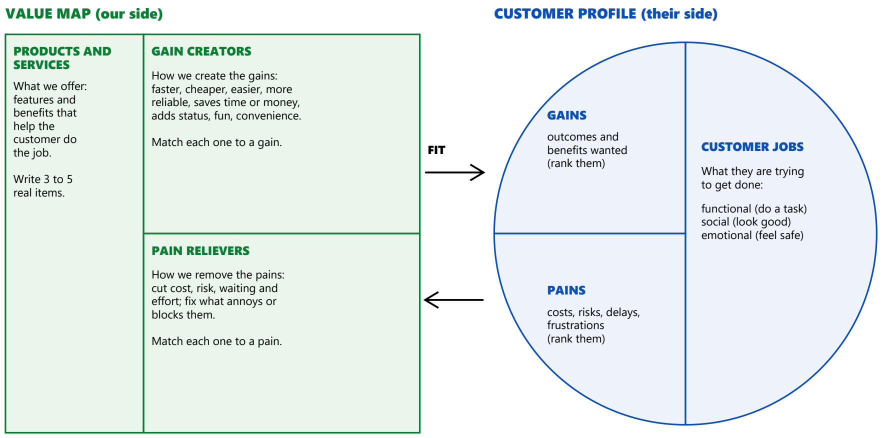
Main sections of a business plan. Executive summary · general company description (mission, vision, goals, ownership) · products and services · marketing plan · operational plan · management and organisation · start-up expenses and capitalisation · financial plan (sales forecast, profit and loss, balance sheet, cash flow) · appendices.

The three tests every business plan must pass. **1 Reality test:** is there a real market and can the product or service be built and delivered? (market opportunity and technical feasibility; evidence: market research, customer interviews, prototypes, market size). **2 Competitive test:** can the venture build a sustainable competitive advantage against competitors and barriers to entry? (evidence: competitor analysis, unique value proposition, unfair advantage, niche). **3 Value test:** is the business financially attractive for the owners and investors? (evidence: sales forecast, profit and loss, cash flow, break-even, funding required, return on investment, exit). To justify your own plan, point to the sections: market research and customer analysis (reality), competitive analysis and value proposition (competitive), financial plan and cash flow (value).

Investors, pitching and what they investigate. **Bankers** want orderly repayment (loan amount, use, repayment terms, collateral). **Investors** want dramatic growth and a share of the rewards: funds needed now and in 2 to 5 years, how the funds will be used, estimated return on investment, the **exit strategy** (buyback, sale, IPO), percentage of ownership given up, milestones, reporting and board involvement. **Four things to emphasise when pitching to venture capitalists or angels:** (1) **the team** (experience and commitment); (2) **problem, solution and market opportunity** (size, growth, customer pain); (3) **traction and business model** (customers, revenue, unit economics, competitive advantage); (4) **financials, the ask and the exit** (use of funds, ROI, milestones); for each, explain why it matters to the investor's decision and risk. **Areas investors investigate before investing:** the management team and track record · market size and demand · product, technology and intellectual property · competition and competitive advantage · business model and financial projections · legal and regulatory position · exit options; the reason is that checking them reduces the investor's risk and estimates the return. **Pitch deck essentials:** problem, solution, product demo, market size, business model, traction, competition and advantage, go-to-market, team, financials and the ask, vision and exit. **Funding stages:** bootstrapping → angel investors (individuals or angel networks) → venture capital → growth rounds; **incubators** support early ideas (space, mentoring); **accelerators** run fixed-term programmes for faster growth, often for equity; **crowdsourcing** gets ideas, funding or work from a crowd online.

SAMPLE PAPER 2025 STYLE: CANVAS TEMPLATES AND COMMON SAMPLE ENTRIES

Value proposition canvas (Q2 part 1): draw both shapes, fill them, then show the fit



VALUE MAP (our side)

PRODUCTS AND SERVICES
What we offer: features and benefits that help the customer do the job.
Write 3 to 5 real items.

GAIN CREATORS
How we create the gains: faster, cheaper, easier, more reliable, saves time or money, adds status, fun, convenience.
Match each one to a gain.

PAIN RELIEVERS
How we remove the pains: cut cost, risk, waiting and effort; fix what annoys or blocks them.
Match each one to a pain.

FIT

CUSTOMER PROFILE (their side)

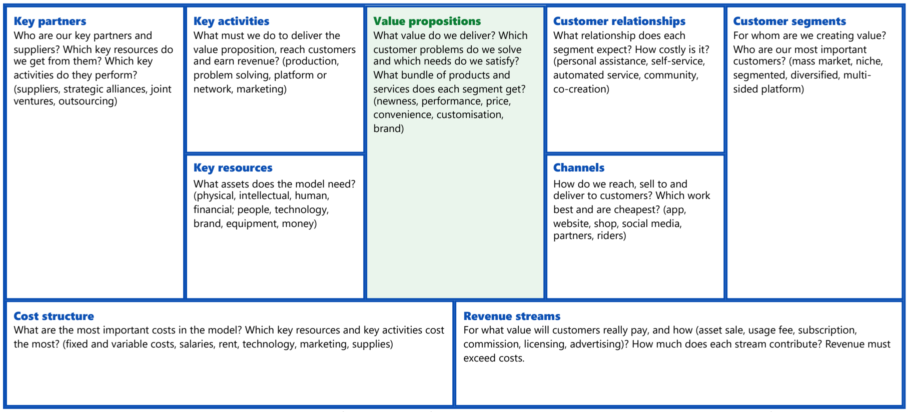
GAINS
outcomes and benefits wanted (rank them)

CUSTOMER JOBS
What they are trying to get done:
functional (do a task)
social (look good)
emotional (feel safe)

PAINS
costs, risks, delays, frustrations (rank them)

Order: 1 customer segment named in the scenario, 2 jobs, 3 pains, 4 gains, 5 products and services, 6 pain relievers, 7 gain creators. Fit = every pain and gain has an answer.

Business model canvas (Q2 part 2): fill all nine blocks and tie each one back to the value proposition



Key partners
Who are our key partners and suppliers? Which key resources do we get from them? Which key activities do they perform? (suppliers, strategic alliances, joint ventures, outsourcing)

Key activities
What must we do to deliver the value proposition, reach customers and earn revenue? (production, problem solving, platform or network, marketing)

Key resources
What assets does the model need? (physical, intellectual, human, financial; people, technology, brand, equipment, money)

Value propositions
What value do we deliver? Which customer problems do we solve and which needs do we satisfy? What bundle of products and services does each segment get? (newness, performance, price, convenience, customisation, brand)

Customer relationships
What relationship does each segment expect? How costly is it? (personal assistance, self-service, automated service, community, co-creation)

Channels
How do we reach, sell to and deliver to customers? Which work best and are cheapest? (app, website, shop, social media, partners, riders)

Customer segments
For whom are we creating value? Who are our most important customers? (mass market, niche, segmented, diversified, multi-sided platform)

Cost structure
What are the most important costs in the model? Which key resources and key activities cost the most? (fixed and variable costs, salaries, rent, technology, marketing, supplies)

Revenue streams
For what value will customers really pay, and how (asset sale, usage fee, subscription, commission, licensing, advertising)? How much does each stream contribute? Revenue must exceed costs.

Layout to copy: customer side on the right (segments, relationships, channels), infrastructure on the left (partners, activities, resources), value proposition in the middle, costs bottom left and revenue bottom right.

Common sample entries: pick what fits the scenario and adapt it

VALUE PROPOSITION CANVAS (CUSTOMER PROFILE, THEN VALUE MAP)	
Customer jobs	do a task (order food, book a stay, pay a bill, find a service) · learn or get information · save money · stay safe and healthy · look good to others · feel in control
Pains	too expensive · slow, long waiting · hard or confusing to use · unreliable or poor quality · risk of fraud or being cheated · no trust in sellers · wasted time and effort
Gains	saves time and money · convenience · good quality and freshness · trust and safety · easy to use · wide choice · status or recognition · a good experience
Products and services	mobile app or website · delivery or booking service · subscription plan · customer support · secure online payment · loyalty programme
Pain relievers	lower or transparent prices · fast delivery · simple app · verified providers · ratings and reviews · refund guarantee · 24/7 support
Gain creators	discounts and rewards · personalised offers · more choice · live tracking · free delivery · eco-friendly options · saves time
BUSINESS MODEL CANVAS (NINE BLOCKS)	
Customer segments	students · working professionals · families · small businesses · tourists · elderly people · a two-sided platform (customers and providers such as farmers, hosts, sellers)
Value propositions	cheaper · faster · easier and more convenient · higher quality or fresher · trusted · customised · one place for everything
Channels	mobile app · website · social media (Facebook, Instagram, TikTok) · WhatsApp · own shop or kiosk · delivery riders · partner outlets · word of mouth
Customer relationships	self-service · automated (app notifications, email) · personal support (chat, hotline) · community and reviews · loyalty rewards · feedback and co-creation
Revenue streams	product sales · delivery fee · subscription or membership · commission per order or booking · advertising · premium listing · licensing
Key resources	app and servers · development team · drivers and vehicles · supplier network · brand · capital · customer data · equipment and premises
Key activities	building and maintaining the platform · sourcing and quality checking · logistics and delivery · marketing · customer support · payment processing
Key partners	suppliers and farmers · payment gateway · delivery companies · banks · cloud provider · regulators · marketing partners
Cost structure	salaries · technology and hosting · delivery and transport · marketing · supplier costs · rent · licences and insurance · payment fees